University of Science and Technology of China
Department of Chemical Physics, Class of 2026

XIANGYU XIE

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EDUCATION

Hefei, China | **University of Science and Technology of China** | **Fall 2022 – Expected Spring 2026**

- **B.S. in Chemical Physics**, Class of 2026
- **GPA:** 3.23/4.3 – Demonstrating consistent upward trajectory: Freshman (2.6), Sophomore (3.4), Junior (3.5)
- **Advisor:** Prof. Jun Jiang (Computational Chemistry & AI for Science)
- **Relevant Coursework:** Quantum Mechanics, Statistical Mechanics, Mathematical Physics Equations, Advanced Quantum Mechanics, Quantum Chemistry, Linear Algebra

RESEARCH EXPERIENCE

Undergraduate Thesis | **Supervisor: Prof. Jun Jiang** | **Fall 2025 – Spring 2026**

Prediction of Room-Temperature Phosphorescence Molecular Properties Based on Equivariant Graph Neural Networks

- **Dataset & Targets:** Built a 3D molecular graph pipeline for about 6,000 organic RTP candidate molecules with quantum-chemistry labels for $\Delta E_{S_1-T_1}$, $SOC_{T_1\to S_1}$, $SOC_{T_1\to S_0}$, and f_{osc,S_1}
- **Model Architecture:** Designed and implemented an 8-layer EGNN_Sparse_Network using 21-dim atom node features, 6-dim bond edge features, 3D coordinates, and equivariant message passing
- **Baseline & Training:** Compared against a character-level SMILES Transformer baseline under the same 4,800/1,050/150 train/validation/test split; applied rotation/translation augmentation only during EGNN training
- **Results:** Achieved test-set R^2 values of 0.8781, 0.6263, 0.9855, and 0.7844 for $\Delta E_{S_1-T_1}$, $SOC_{T_1\to S_1}$, $SOC_{T_1\to S_0}$, and f_{osc,S_1}, respectively
- **Conclusion:** Showed that explicit 3D geometry and equivariant inductive bias improved isolated-molecule RTP excited-state property prediction over the sequence baseline
- **Skills:** PyTorch, Graph Neural Networks, SE(3) Equivariance, Quantum Chemistry, Python, Linux

Undergraduate Research | **Supervisor: Prof. Linsong Cui** | **Fall 2023 – Fall 2024**

Design and Synthesis of Light-Emitting Materials Based on Doublet-Emissive Mechanism

- **Project:** Undergraduate Research Program; successfully completed and passed final evaluation
- **Molecular Design & Synthesis:** Designed TTM-AnQ radical by coupling TTM radical with 9,10-anthraquinone acceptor; executed multi-step synthesis via Friedel–Crafts, Suzuki coupling, and oxidation
- **Characterization:** Conducted UV-Vis, fluorescence, CV, and transient fluorescence measurements; observed emission red-shift from 580nm to 695nm upon acceptor incorporation
- **Key Finding:** Demonstrated SOMO energy modulation (-2.82eV → -3.18eV), suggesting an emission-tuning pathway beyond traditional donor-functionalization; conducted an exploratory undergraduate study of acceptor-functionalized radical emitters

Undergraduate Competition | **Digital Intelligence Track** | **Fall 2024 – Spring 2025**

Multiscale Molecular Simulation-Driven Physical Chemistry Experiment Teaching Innovation

- **Framework:** Developed ”Experiment + Computation” integrated protocol combining MD simulations (GROMACS) with DFT calculations (Gaussian 16, B3LYP/6-31G+g(d,p))
- **Simulations:** Built n-butanol/cyclohexane models (17:400 and 1:400); performed energy minimization, equilibrium, and production runs at 30°C; extracted 100 trajectory frames for DFT computation
- **Results:** Achieved 1.78D predicted dipole moment (6.0% deviation from experimental 1.66D); revealed microscopic origin of infinite dilution extrapolation method

- **Award: First Prize, Central China Division** of the 5th National Undergraduate Chemistry Experiment Innovation Design Competition (Digital Intelligence Track)

Research Demo **Spring 2025 – Spring 2026**

MultiSpec-GeoDiff: Spectra-Driven Molecular Structure Inversion

- **Project:** Built a runnable Stage-I feasibility loop for spectra-driven molecular structure inversion using 200 experimental NIST IR spectra with coordinate-aware spectral encoding and spectral-distance-bias attention
- **Retrieval & Reranking:** Developed candidate retrieval, forward-spectrum reranking, Top-K visualization, traceable outputs, and CI-tested reproducible demo workflow

Awards and Honors

- **USTC Outstanding Student Scholarship** – Awarded for academic excellence (2024, 2025)
- **USTC Strengthening Basic Disciplines Plan Grant** – Top 20% of students eligible (2025)
- **Undergraduate Competition, Central China Division – First Prize** – 5th National Undergraduate Chemistry Experiment Innovation Design Competition, Digital Intelligence Track (2025)

Academic Exchanges

Research Visits **Multiple Institutes**

- **August 2023:** Dalian Institute of Chemical Physics, Changchun Institute of Applied Chemistry, Shenyang Institute of Metal Research
- **August 2025:** Shanghai Institute of Organic Chemistry, Shanghai Institute of Ceramics, Suzhou Institute of Nano-Tech and Nano-Bionics
- **July 2025:** Hong Kong University, HKUST, Chinese University of Hong Kong, City University of Hong Kong

Teaching Experience

Teaching Assistant

- **Fall 2024:** Complex Functions (for Chemistry) – Led discussion sections, graded assignments, provided one-on-one tutoring
- **Spring 2025:** Mathematical Physics Equations (B) – Facilitated problem-solving sessions, developed supplementary materials
- **Spring 2026:** Quantum Physics – Demonstrating strong command of theoretical foundations

Technical Skills

Programming and Tools

- **Machine Learning / Geometric Deep Learning:** PyTorch, Graph Neural Networks, EGNN, Tensor Field Network, SE(3)-Transformer, Transformer architectures
- **Programming & AI-assisted Development:** Python; Claude Code, Cursor, Codex, OpenCode; agentic coding and research-oriented prototyping workflows
- **Computational Chemistry:** DFT/TD-DFT calculations, Quantum Chemistry Software (Gaussian, VASP), Molecular Visualization
- **Mathematical Foundations:** Linear Algebra, Group Theory, Probability and Statistics, Complex Functions
- **General Tools:** LaTeX, Git, Linux/Unix command line, Data Analysis (numpy, pandas, matplotlib)

Strengths

- **Solid Mathematical Foundation:** Strong grasp of theoretical chemistry and mathematical physics principles
- **Analytical & Professional Competence:** Derive novel research questions, formulate rigorous analytical approaches, and execute complex research projects from conception to completion
- **Interdisciplinary & Collaborative:** Combine wet lab synthesis with computational modeling; experience working across multiple research groups and institutions
- **First-principles reasoning and AI4S modeling:** Analyze scientific problems from physical and mathematical foundations; understand geometric, equivariant, and Transformer-based deep learning models for AI for Science.